Messy master data hides money. Tesseract fixes it at the front door.

TEAM CTFUDDERS
SMU DSA 2026 FINALS

LIVES GROUP
5 COUNTRIES, ONE SAP BUSINESS ONE

RUNS ON MICROSOFT 365
LANDS IN SAP BUSINESS ONE

SYNTHETIC, ILLUSTRATIVE DATA
ALL NAMES FICTIONAL

Tesseract

Amy needed one item code. It took 18 days.

Day

18

Day 1	New item code request: K2EDTA blood collection tubes
Day 3	RE: New item code request: K2EDTA blood collection tubes
Day 7	RE: RE: FW: new item code, licence no. missing?
Day 12	FW: RE: RE: FW: new item code, still waiting on QA/RA sign-off



2 days	waiting for MDM
5 days	bounced: missing QA/RA field
9 days	3 approvers, one at a time
2 days	MDM queue + manual keying

Day 18: she tries to raise a sales order. The item still isn't in SAP.

S$1.34M of customer orders waited on a master-data setup in the last 12 months.

Amy's case: case brief, App C. Day split: our reading of it. S$ figure: synthetic, illustrative data.

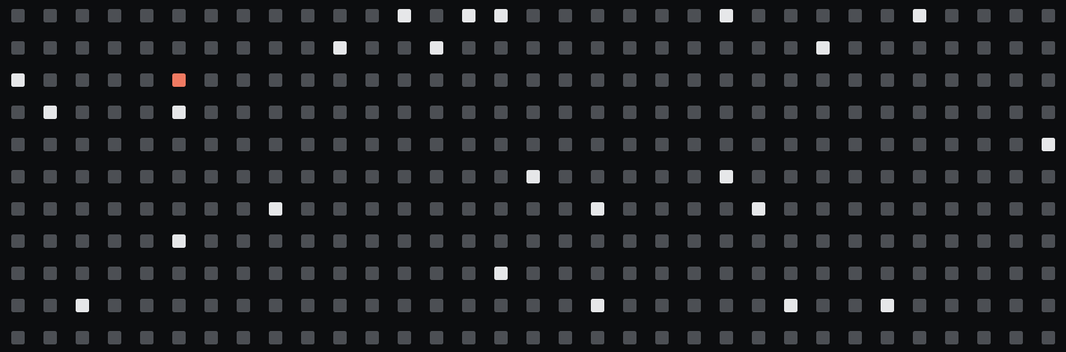
In SAP, one hospital can look like three different customers.

SAP code	Name as keyed	Entity	Rank in Malaysia	Last 12 months
C-MY-00061	Specialist Hospital Bukit Damai	QTS	#7	Flat, +4%
C-MY-00029	Bukit Damai Specialist Hospital	SPD	#10	Lost customer, −66%
C-MY-00091	Bukit Damai Spec Hospital	SPD	#11	New customer

Raw SAP records, Malaysia, Oct 2025–Sep 2026 vs the 12 months before. Synthetic, illustrative data.

Merge the duplicates, and Malaysia's #2 account is down 32%.

SAP code	Customer	Entity	Rank in Malaysia	Last 12 months
G-MY-0002	**Bukit Damai Specialist Hospital** 3 SAP records merged C-MY-00029 C-MY-00061 C-MY-00091	SPD + QTS	**#2**	**−32%** S$1.59M → S$1.08M



388 customer records in SAP

363 distinct customers

a customer that had duplicates Bukit Damai Specialist Hospital

25 of 388 records (6.4%) are duplicates. Merging moves no country total; it changes who you call.

Synthetic, illustrative data.

Every symptom in the brief starts in the same place: master data is requested by email.

Today (brief §3–4, App B)	So	How big
Free-text email, no standard form	Requests arrive incomplete and bounce	**40%** bounce at least once
No way to check what already exists	Duplicates slip into SAP	**6%** of new customers (A13)
Forwarded to approvers one at a time, no reminders	Nobody knows where a request is	Median **14 days** to set up
Bank details and credit terms in ordinary email	No trustworthy record of who approved what	no audit trail
A new business unit means redrafting templates, with IT	Marcus's team's requests sat for two weeks	**2 weeks** idle

And the regional numbers are built on these same records.

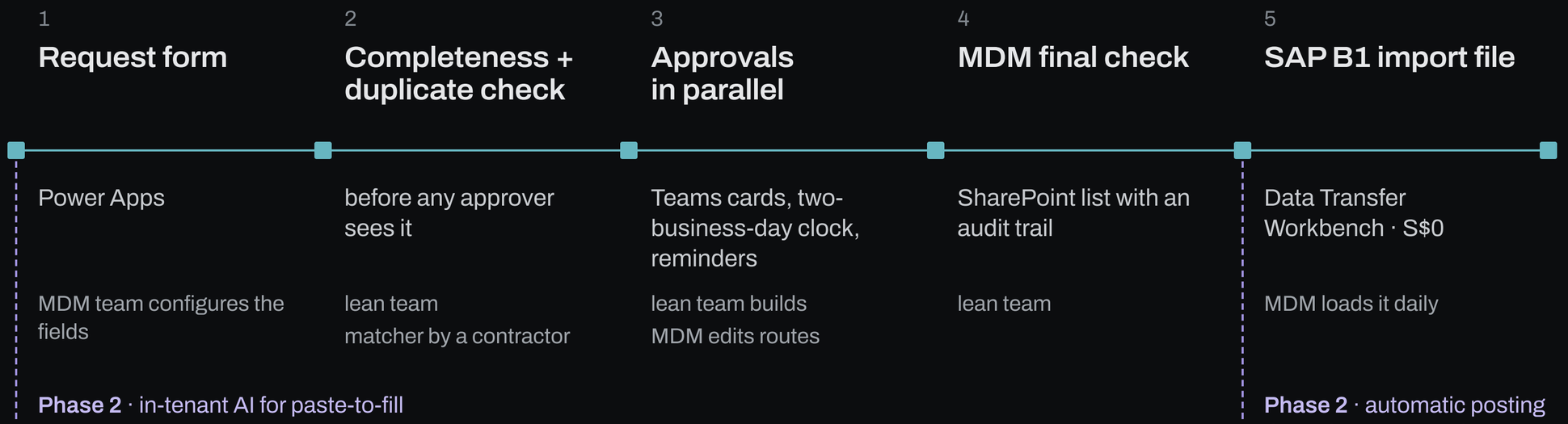
40%, 6% and 14 days are labelled assumptions (A13, A26), illustrative. Marcus's two weeks: case brief, App B.

A good fix uses what Lives already owns and stops bad data before anyone approves it.

1	**Build on what Lives already owns.**	Microsoft 365 is the front door; SAP Business One stays the system of record. SAP Business One's built-in approvals are document-centric.
2	**One record per real customer.**	Check for duplicates before approval; report on de-duplicated customers.
3	**Fix the process before buying software.**	Approvers decide in parallel, in Teams, on a two-business-day clock.
4	**The MDM team owns the rules.**	A new business unit, field or approval step without an IT ticket.

Checked against a better email template, SAP B1 approvals, Teams Approvals alone, Power Apps alone and an MDM suite (appendix A1).

Tesseract is one front door on Microsoft 365, and every approved record lands in SAP Business One.



No new licences in phase 1 | **Lean team of 2 · 6 weeks** | **≈15 days of outside help**

In five minutes, you'll watch one request travel from Amy's email to a decision in Regional Pulse.

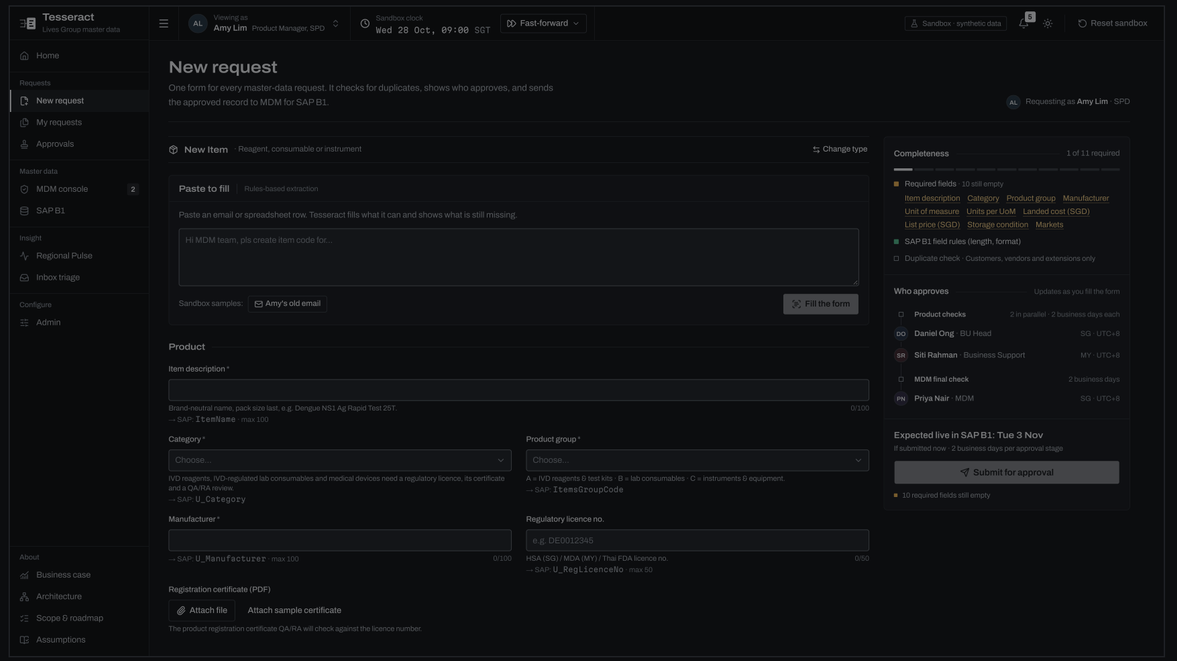
1 **Amy**
pastes her old email; the form fills and routes itself

2 **Marcus**
is stopped from creating a duplicate customer

3 **Wei Jie**
approves in Teams; credit terms never leave Tesseract

4 **Priya**
sends it to SAP; S$36,000 of waiting orders are released

5 **Regional Pulse**
merge the duplicates, and the hidden decline appears



The live app, in this browser

Live prototype · synthetic data · Teams, Outlook, SAP B1 and the clock are simulated in the browser.

Once customers are merged, Pulse tells Malaysia to call Bukit Damai by 13 Nov.

Growth	**S$53.4M, +4.0%** +7.6% Jul–Sep	Same 12 months a year apart, de-duplicated customers
Forecast	**11 → 9 per 100**	Units off per 100 ordered · wins 7 of 10 cells
Stuck demand	**S$345,800 → S$309,800**	Orders waiting on a setup · –S$36,000 you just watched
Action	**S$620k addressable** before conversion	Owner: BU Head, SPD Malaysia · one plan across SPD + QTS

Synthetic, illustrative data · Growth = PO value (Oct 2025–Sep 2026) ÷ PO value (Oct 2024–Sep 2025) – 1 · Forecast: Products A + B, on 6 months the model never saw.

Median setup time drops from 14 days to 4 business days or less.



≈1,750 hours a year back

1,751 h, ≈0.94 FTE: capacity, not headcount

≈43% of the time saved is a process change

Approving in parallel needs no new software

Median 14 days: assumption A26. Lanes are schematic, in the proportions of Amy's 18 days (slide 2). Hours at 80 requests a month; from labelled assumptions (A12–A26).

Phase 1 costs S$39.9k with no new licences, and pays back in about six months.

Requests a month	Hours back a year	Net benefit a year	Payback	3-year net
40	1,115	S$49.8k	9.6 months	S$109k
80 (base)	**1,751**	**S$81.5k**	**5.9 months**	**S$205k**
120	2,386	S$113.3k	4.2 months	S$300k

Phase 1: S$39.9k one-off + S$6.0k a year to run

Lean team of 2 for 6 weeks, a 15-day contractor, MDM set-up, 30-minute onboarding. Cash ≈ S$9,750 (the contractor); the rest is internal time. No new licences for the workflow.

▫ Phase 2, only after the pilot

S$22.5k + ≤ S$2.6k a year in licences. Automatic posting saves only the import step: about a 19-year payback on hours, so it is a speed decision.

Hard hours only, at S$50/h. Approver time, grants and stuck-demand revenue are not counted. From labelled assumptions (A12–A26).

We prove it in a six-week pilot, decided at a week-6 gate.

LENGTH	COUNTRIES	DOMAINS	BUSINESS UNITS	WEEK 1
6 WEEKS	SINGAPORE + MALAYSIA	CUSTOMERS + ITEMS	SPD + QTS	RERUN THE ANALYSIS ON REAL SAP B1 EXTRACTS

KPI	Today	Target by week 6
Requests through Tesseract	0% (email)	**≥ 90%**
Median setup time	14 days	**≤ 4 business days**
Requests bounced for missing information	40%	**< 15%**
Duplicates stopped at intake	not measured	**counted weekly**
Customer orders waiting on a setup (S$)	tracked live	**falling**

Week-6 gate

Median over 6 business days, or adoption under 70% → **fix or stop**

Targets met

→ Thailand, Vietnam, Myanmar, then vendors and cost centres

Baselines are labelled assumptions (A13, A26).

The biggest risk is habit, so we take away the email path instead of asking people to change.

Risk	What we do	What we watch weekly
People keep emailing	After 4 weeks the MDM mailbox auto-replies with the form link; a champion per entity and country; 30-minute onboarding	Share of requests through Tesseract
Approvers miss one more notification	One-click approval in Teams; reminder after 1 business day, escalation after 3	SLA scoreboard: who is waiting on whom
Sensitive values leak	Bank and credit values never on the card; Finance and MDM only; every view logged; nobody approves their own request	Audit trail
Our numbers don't hold on real data	Week-1 rerun on SAP B1 extracts; the matcher suggests, MDM decides	The week-6 gate

Our ask: approve phase 1 (S$39.9k, mostly internal time, no new licences for the workflow) and a six-week pilot in Singapore and Malaysia, decided at the week-6 gate.

≤ 4 business days
median setup time, from 14 days

≈1,750 hours
back a year

≈6 months
payback

LIVE PROTOTYPE
onerecord-lives-dsa26.vercel.app

CODE
github.com/ethantyh03/onerecord

DEMO VIDEO
youtu.be/5tieuSmd2Aw

TEAM CTFUDDERS
SMU DSA 2026

One hospital. One record.

From labelled assumptions (A12–A26); synthetic, illustrative data.

SAP Business One's built-in approvals are document-centric, and the simpler options fix only half the problem.

Option	One-off	Stops incomplete	Stops duplicates	Fast + visible approvals	MDM changes it without IT	Regional analytics
Better email template + shared tracker	~S$0	~	✗	✗	✓	✗
SAP B1 approval procedures	S$0	✗	✗	—	—	✗
Teams Approvals only	~S$0	~	✗	✓	✓	✗
Power Apps + SharePoint + Power Automate only	~S$30k	✓	~ exact match only	✓	✓	✗
Buy an MDM suite / B1 add-on	High (not verified) + IT	✓	✓	✓	~	~
Tesseract	**~S$40k, no new licences for the workflow**	✓	✓	✓	✓	✓

✓ yes · ~ partly · ✗ no · — not applicable

SAP B1 approval procedures cover sales, purchasing, inventory and payment documents (SAP Learning, “Implementing the approval processes in SAP Business One”).

Phase 1 is configuration on Microsoft 365, built by a lean team of two, with no new licences for the workflow.

Part	In production	Licence	Who builds it
Request form	Power Apps canvas app, default environment, no Dataverse	Existing Microsoft 365	Lean team; MDM configures fields
Store	SharePoint list, plus a restricted list for bank and credit fields	Existing	Lean team
Routing + approvals	Power Automate, standard Approvals connector → Teams cards / Outlook	Standard connector	Lean team; MDM edits routes
Duplicate check	Nightly job writes match keys to a SharePoint list; the form looks them up	Existing	Contractor, ~15 days
Audit	SharePoint version history + Purview logs + a hash column	Existing	Lean team
Landing in SAP B1	Import file via SAP's Data Transfer Workbench, loaded daily	S$0, ships with SAP B1	MDM team, no build
Regional Pulse	Power BI on SAP B1 extracts	Power BI Pro for ~10 viewers (≈USD 14/user/month, approx.), only if not already owned	Lean team analyst
Phase 2 posting	Process licence + 1 Premium maker → Service Layer	≤ USD 1,980 a year (≈S$2,633); pay-as-you-go is cheaper below ~250 runs a month	SAP partner, ~15 days

This web app is the demo vehicle. The repo ships the production kit: Teams card JSON, the flow spec, import templates and the SharePoint schema.

Payback counts hard hours only: 1.25 hours saved per request, valued at S$50 an hour.

Hours saved per request: rework 0.40 (40% bounce × 1.0 h) + forwarding and chasing 0.50 + keying vs import file 0.35	**1.25 h**
80 requests × 1.25 h + ≈1.47 duplicates stopped × 4 h	**105.9 h a month**
+ regional consolidation, 5 analysts × 1 day	**40 h a month**
= 1,751 h a year × S$50 = S$87,528 – S$5,985 run cost	**S$81,543 net a year**
S$39,900 ÷ (S$81,543 ÷ 12); 8.3 months on workflow hours alone	**5.9 months**

Not counted: approver time (S$85–110/h), stuck-demand revenue, an EDGE grant of ≈S$16k if Lives qualifies as an SME.

S$50/h is derived from the MOM 2025 median wage plus bonus and CPF ÷ 1,872 h. All inputs are labelled assumptions (A12–A26).

The duplicate check catches 87 of 100 duplicates in countries it never saw, and 93 of 100 warnings are real.

1

Clean the name: case, punctuation, legal forms such as Sdn Bhd, abbreviations such as Natl / Univ / Hosp

2

Score it: the better of Jaro-Winkler and token-set similarity

3

The same tax ID lifts the score to 97%; the same postcode, phone or email domain adds 3 points each

4

MDM decides

How we tested it: threshold tuned on Singapore + Malaysia, frozen, then scored once on Thailand, Vietnam and Myanmar.

Precision 0.929

93 of 100 warnings are real

Recall 0.867

87 of 100 duplicates caught

13 found · 2 missed · 1 false alarm

Held-out countries only

Weakest cases, shown: romanisation variants and blank tax IDs (6 of 8 caught each).

Synthetic, illustrative data.

All data is synthetic, and the same scripts run unchanged on Lives's SAP extracts in week 1.

One seeded script	Seed 42; 36 months, Oct 2023–Sep 2026; every parameter published; every planted pattern listed and checked (a hidden decline behind duplicates; orders booked to the wrong country; a false cross-sell gap).
What a good score proves	The pipeline can find these patterns, not that they exist at Lives.
Week 1 of the pilot	Rerun on SAP B1 extracts: orders (`ORDR`/`RDR1`), quotations (`OQUT`), business partners (`OCRD`), items (`OITM`).
Names	People and hospitals are fictional; the brief's example hospital appears only as a master record, with no sales. Amy's 18 days is from the brief; the day split is ours.

Our seasonal forecast beats today's rolling average on months it never saw, and we show where it loses.

Today vs ours	Today: rolling 12-month average. Ours: a damped seasonal model (Holt-Winters) per country × product, Products A and B.
Test fixed in advance	6 starting points (end of Mar to end of Aug 2026), 1–3 months ahead, refitted each time on past data only.
Result	Off by **11 → 9 units per 100 ordered**; wins 7 of 10 country × product cells; loses MY·A, TH·B and MM·B.
Honest limits	The gain is seasonal (off-season it is +0.2 points). With seasonality switched off our advantage is −0.6 points (≈ 0), so it doesn't invent signal; another seed with half the dengue effect still wins by 2.3 points. **On uncleaned records ours is worse (15 vs 13 per 100): clean first, then model.** Product C (instruments) is planned from tenders, not forecast.

Synthetic, illustrative data.

Sensitive values never travel in a message, and nobody approves their own request, even after an escalation.

Bank and credit

The Teams card shows “•••• · view in Tesseract (Finance and MDM only)”. In production: a restricted SharePoint list with item-level permissions through Entra ID (SharePoint has no column-level security, hence a separate list). Every view is logged. Vendor bank details need a call-back to an independently sourced number, logged.

Segregation of duties

The requestor and their delegate can't approve; nobody holds or decides two steps of one request (so the Finance approver can't be the MDM keyer); any bank field, or a credit limit over S$50k, adds a Finance Director step.

When nobody acts

A reminder after 1 business day; after 3, the step escalates to the approver's delegate, else their manager, skipping anyone who raised the request or holds another step on it; if nobody is eligible, MDM is told. Out-of-office goes straight to the delegate.

Audit

Every view and decision on a SHA-256 hash chain; any edit after the fact breaks it.

In the prototype, masking is enforced in the interface only, and the screens say so.

We land records through SAP's own import tool, and we left six things out on purpose.

Import file now, Service Layer later.

The import file keeps MDM's human check before anything enters SAP. Automatic posting saves only the load step, ≈0.15 h a request: 12 h a month at 80 requests = S$7,200 a year, minus S$2,633 licences and S$3,375 maintenance = **S$1,192 net a year → the S$22.5k build pays back in ~19 years** (~7.4 on pay-as-you-go; ~4.7 at 120 requests a month). The Service Layer payload is already generated and validated, marked “not sent”. Extending a customer is one update to the shared business-partner record, never a second record.

Not in version 1, by design

- automatic posting to SAP
- merging existing SAP duplicates (a separate stewardship project; v1 uses a read-only crosswalk)
- our own security model (Microsoft 365 and Entra ID provide it)
- a drag-and-drop workflow designer
- deep-learning forecasts (not justified on monthly data at this scale)
- change requests other than Extend

The prototype also runs no live AI model: rules plus cached examples, badged; in-tenant AI is phase 2.